



ROBOGRINDER AT VIRGINIA TECH

On the Grind Since 2017

INFORMATION PACKET
2026 - 2027

About Us

Founded in 2017, RoboGrinder is a student-led engineering team recognized by Virginia Tech's Department of Mechanical Engineering and the Bradley Department of Electrical and Computer Engineering. The team works under the advisement of Dr. Dylan Losey (ME) and Dr. Tyler Milburn (ECE), building on the long-time guidance of our previous ECE advisor, Dr. William T. Baumann. Over the past decade, we have established ourselves as one of the premier collegiate robotics teams in the United States, with 2027 marking our 10th anniversary.

We proudly co-founded the original RoboMaster North America conference alongside UCSD Triton Robotics in 2021, establishing the foundation for what has evolved into the ARC Robotics Competition (ARCC)—widely recognized as one of the most technically demanding student robotics challenges in the world.

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Our Mission

To empower students to apply their engineering knowledge from the classroom towards solving complex mechanical, electrical, and programming problems. Intercollegiate competitions, such as the ARC Robotics Competition (formerly known as RoboMaster North America), provide unique challenges that require various engineering disciplines to collaborate to not only win, but to build effective and efficient robots—training the next generation of industry-ready innovators.



Our Track Record

2026

2ND in 3v3

ARC Robotics Competition Season One

2025

1ST in 3v3

RoboMaster University League North America

2024

2ND in 3v3

RoboMaster University League North America

2023

4TH in 3v3, 4TH in 1v1

RoboMaster University League North America

2022

1ST Prize in ICRA AI Challenge

2ND in 1v1

RoboMaster University League North America

2021

2ND in 3v3

RoboMaster University League North America

2019

2ND Team in International Group

1ST Team in North America

2ND Prize in Final Tournament

Ranked Global Top 32 University Teams

RoboMaster University League

2018

1ST Prize in International Regionals

2ND Prize in Final Tournament

RoboMaster University League

2017

Founding Year

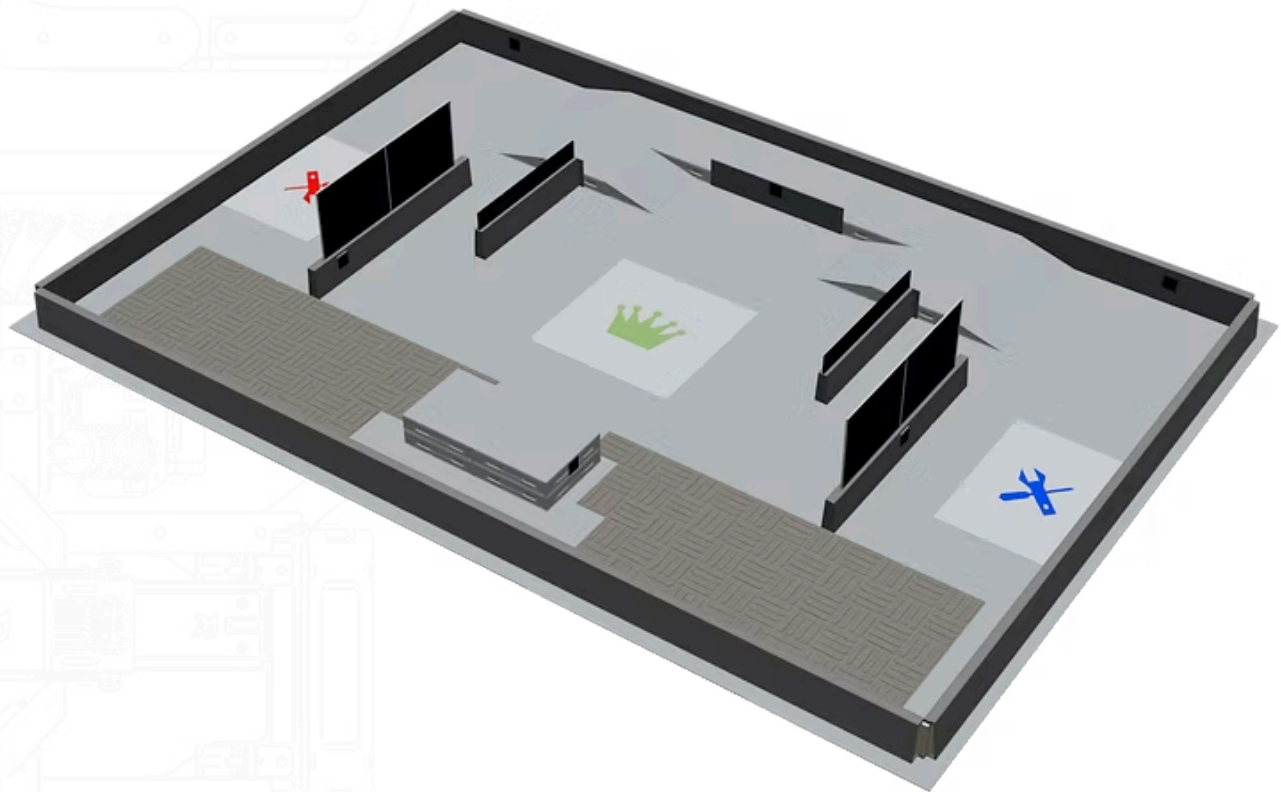
3RD Prize in Final Tournament

RoboMaster University League

About The Competition

The ARC Robotics Competition (formerly RoboMaster North America) plays like a real-world crossover between a First-Person Shooter (FPS) and a Multiplayer Online Battle Arena (MOBA). Two teams face off in high-speed tactical matches, deploying fleets of custom-built, semi-autonomous, and fully autonomous robots. Operators control these machines with no direct line of sight to the robots, steering and aiming entirely through real-time FPV camera feeds.

To win, teams must coordinate different robot classes to capture strategic map objectives and defeat the opposing base. Hits are tracked instantly by pressure-sensitive armor plates connected to a digital referee system, which monitors hit points (HP) and electrical power limits in real time.

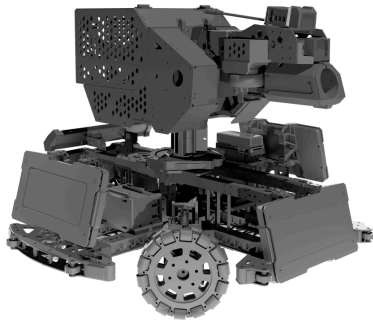


ARC Robotics Competition Season One 3v3 Arena

Our Robots

The Sentry

A fully autonomous unit utilizing onboard SLAM, NAV2, LiDAR, and computer-vision tracking algorithms to deal damage to enemies without human operators.



The Hero

Heavy, high-torque assault units firing 42mm golf-ball-sized projectiles, designed to deliver massive damage to enemies and fortifications.



The Infantry

Highly agile ground units equipped with 17mm projectile launchers, built to lead the offensive line and duel enemy forces.

The Engineer

A non-combat utility unit engineered with custom grabbing mechanisms to gather resource minerals and revive eliminated teammates on the fly. It is also the most mechanically complex robot in the fleet.

Our Teams

The team is divided into four sub-teams, each owning a different part of the robots and their operation. They work independently but stay tightly coordinated.

Electrical & Embedded

Covers two closely related branches under one umbrella. The electrical side handles the hardware, designing the robots' power distribution, wiring, motor control, and custom PCBs. The embedded side bridges hardware and software by writing firmware in C/C++ that runs on each robot's STM32 microcontroller, controlling its components and managing communication between them.

Mechanical

Designs and builds the physical robots, from chassis, drivetrains, and turrets to firing mechanisms. Working from SOLIDWORKS CAD models through machined and assembled parts, this team turns concepts into machines that hold up in the arena.

Software

Writes higher-level software that controls the robots, including auto-aim, computer vision, autonomous systems, navigation, and simulation. This team makes the robots perceive their surroundings, aim, move, and operate on their own.

Operations

Keeps the team running off the field, handling sponsorships, funding, recruitment, and event logistics. This team also leads the team's media and outreach, including social media, content creation, and public presence that support everyone else's work.

Getting Involved

RoboGrinder welcomes students of all majors and experience levels.

Many members join without prior robotics experience and learn by doing, working alongside more experienced students from their first season on. That said, the team is competitive, and spots are limited, so prior experience and personal projects strengthen an application considerably. Students newer to robotics often start on the Operations team, contributing right away while getting an inside look at how the team works, then move to a technical sub-team when they reapply the following season.

Contact Us

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Socials

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